

01c — Domain Names

Key Concepts

Term	Definition
Domain Name	Human-readable alias that maps to an IP address
TLD	Top-Level Domain — <code>.com</code> , <code>.org</code> , <code>.io</code> — managed by ICANN
SLD	Second-Level Domain — the name you register (e.g., <code>google</code>)
Subdomain	Prefix you control without extra registration (e.g., <code>mail</code> , <code>api</code>)
ICANN	Global body that oversees domain name system
Registrar	Company where you buy/register domains (GoDaddy, Namecheap)
Name Server	Tells the internet where your domain's DNS records live
GeoDNS	Returns different IP for same domain based on user's location
Anycast	Routes user to nearest server using same IP/domain

How It Works

```
https://mail.google.com
      ↑   ↑   ↑
      sub second top-level
      domain domain domain
```

1. You register a domain via a **Registrar** (first-come, first-served)
2. Registrar records your ownership with **ICANN**
3. You point domain to **Name Servers**
4. Name Servers hold **DNS records** mapping domain → IP
5. Browser queries DNS → gets IP → connects to server

Industry Examples

- **Netflix** — uses `api.netflix.com`, `assets.netflix.com` to route traffic to different servers independently
- **Google** — `google.com` returns different IPs based on location (GeoDNS) for nearest server
- **Nike** — ICANN's UDRP process lets trademark holders reclaim misused domains

Interview Questions

Q: What is a domain name? A: A human-readable alias for an IP address, managed via ICANN and purchased through registrars.

Q: What are the parts of a domain name? A: TLD (**.com**), SLD (**google**), and optionally a subdomain (**mail**, **api**).

Q: Who controls domain names globally? A: ICANN oversees the system; registrars sell and manage individual domains.

Q: Can one domain point to multiple IPs? A: Yes — GeoDNS returns different IPs based on user location, routing them to the nearest server.

Q: What happens if two companies want the same domain? A: First-come, first-served. Trademark holders can dispute via ICANN's UDRP process.